EIN3241: Human Factors and Ergonomics

Lecture 8B - Job Design Example

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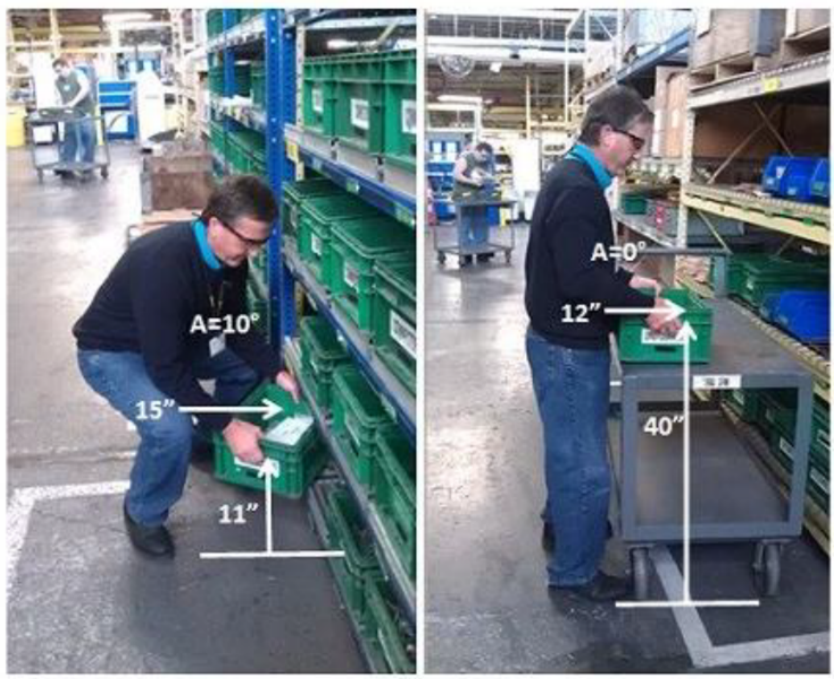
The figure shows a lifting task with two positions: the origin (left photo) and the destination (right photo). The lifting frequency is ≤ 0.2 lifts/min over a 30-minute period, and the average load being lifted is 10 lbs. Perform the NIOSH lifting equation analysis. What RWL will represent the lift? What is the Lifting Index?



Origin Destination

Origin (Left Photo)

- H_0 (horizontal distance) = 15 in
- V_0 (vertical height) = 11 in
- A_0 (asymmetry angle) = 10°

Destination (Right Photo)

- H_d = 12 in
- V_d = 40 in
- A_d = 0°

Additional Parameters

- Load Constant (LC) = 51 lbs
- Frequency Multiplier (FM) = 1.0
- Coupling Multiplier (CM) = 1.0
- Average load = 10 lbs

1. Vertical Travel Distance

$$D = |V_d - V_0| = |40 - 11| = 29 \text{ in}$$

2. Horizontal Multiplier (HM)

Formula: $HM = \frac{10}{H}$

Origin:

$$HM_0 = \frac{10}{15} = 0.67$$

Destination:

$$HM_d = \frac{10}{12} \approx 0.83$$

3. Vertical Multiplier (VM)

Formula: $VM = 1 - 0.0075|V - 30|$

Origin:

$$\begin{aligned} VM_0 &= 1 - 0.0075|11 - 30| \\ &= 1 - 0.0075 \times 19 \\ &= 1 - 0.1425 \\ &= 0.8575 \end{aligned}$$

Destination:

$$\begin{aligned} VM_d &= 1 - 0.0075|40 - 30| \\ &= 1 - 0.0075 \times 10 \\ &= 1 - 0.075 \\ &= 0.925 \end{aligned}$$

4. Distance Multiplier (DM)

Formula: $DM = 0.82 + \frac{1.8}{D}$

With $D = 29$ in:

$$DM = 0.82 + \frac{1.8}{29} \approx 0.82 + 0.0621 \approx 0.8821$$

(Same DM for origin and destination)

5. Asymmetric Multiplier (AM)

Formula: $AM = 1 - 0.0032A$

Origin ($A_0 = 10$):

$$AM_0 = 1 - 0.0032 \times 10 = 1 - 0.032 = 0.968$$

Destination ($A_d = 0$):

$$AM_d = 1 - 0.0032 \times 0 = 1.0$$

6. Compute RWL at Origin

$$RWL_0 = LC \times HM_0 \times VM_0 \times DM \times AM_0 \times FM \times CM$$

$$RWL_0 = 51 \times 0.67 \times 0.8575 \times 0.8821 \times 0.968 \times 1.0 \times 1.0 \approx \boxed{25 \text{ lb}}$$

7. Compute RWL at Destination

$$RWL_d = LC \times HM_d \times VM_d \times DM \times AM_d \times FM \times CM$$

$$RWL_d = 51 \times 0.83 \times 0.925 \times 0.8821 \times 1.0 \times 1.0 \times 1.0 \approx \boxed{34.7 \text{ lb}}$$

8. Final RWL for the Lift

We take the **more conservative (smaller)** of the two RWLs:

- Origin: ≈ 25.0 lb
- Destination: ≈ 34.7 lb

The lift is governed by the **origin**, so:

$$\boxed{\text{RWL for this lift} = 25 \text{ lb}}$$

9. Lifting Index (LI)

Formula: $LI = \frac{\text{Load Weight}}{RWL}$

Origin:

$$LI_0 = \frac{10}{25} = 0.4 \quad (< 1, \text{ acceptable})$$

Since LI value is less than 1, **the lift is acceptable**.